

Lec 06: Performance modeling of computer networks (wrap)

Computer Networks (CSE232) Section A

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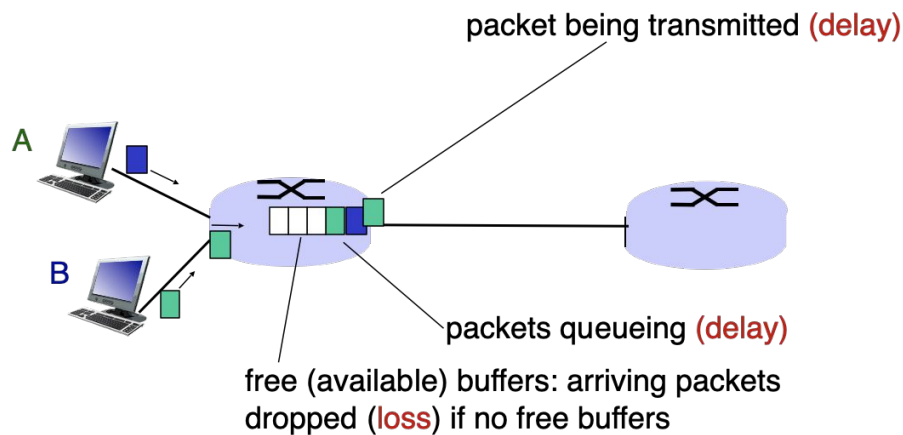
These slides are inspired and adapted from:

1. *Computer Networking: A Top-Down Approach*
6th edition, Jim Kurose, Keith Ross, Pearson, 2017
2. *Course notes by Prof. Mythili Vutukuru, IIT Bombay*
https://www.cse.iitb.ac.in/~mythili/teaching/cs641_autumn2015/index.html



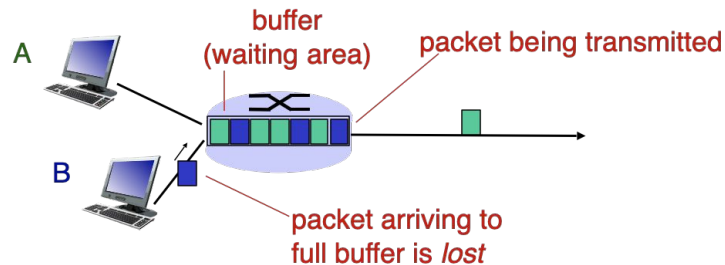
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Packet switching: How do packet delay or packet losses occur?



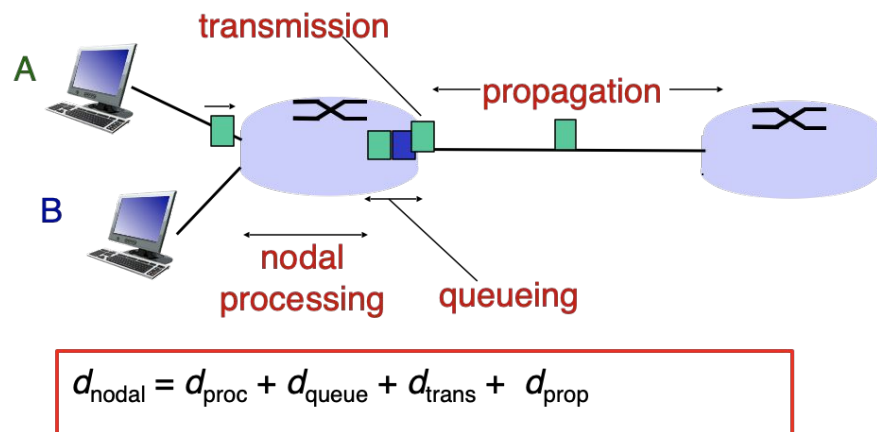
Packet arriving rate > Output link's capacity (temporarily)

- Incoming packet stored in queue (router's buffers) waiting for its turn



- Finite buffer capacity at the router
- Packet arrives when queue is full; packet is dropped
- Lost packet may or may not be retransmitted
 - Retransmission by previous node
 - Retransmission by source

Sources of packet delay



- **Transmission delay (d_{trans})**

- Time required to place the entire packet over the output link
- Depends on link capacity (bandwidth)

- **Propagation delay (d_{prop})**

- Time required for one bit to propagate from one end to another
- Depends on transmission medium characteristics

- **Nodal processing delay (d_{proc})**

- Check bit errors
- Determine output link
- Depends on node specs

- **Queueing delay (d_{queue})**

- Waiting time at output link for congestion
- Depends on congestion level

Sources of packet delay (contd.)

- Transmission delay

- Link data rate = R bits/sec
- Packet size = L bits
- Transmission delay, $d_{\text{trans}} = L/R$ seconds
- Example: Packet size, $L = 1000$ bits, Link data rate = 10^9 bps; $d_{\text{trans}} = 1$ us

- Propagation delay

- Length of the link / distance travelled = d meters
- Speed at which signal travels = s meters/sec
- Propagation delay, $d_{\text{prop}} = d/s$ seconds
- Example: Speed of light, $s = 3 \times 10^8$ m/s; $d = 300$ m; $d_{\text{prop}} = 1$ us

Relation between the link utilization & propagation/transmission delays

$$d_{\text{prop}} < d_{\text{trans}}$$

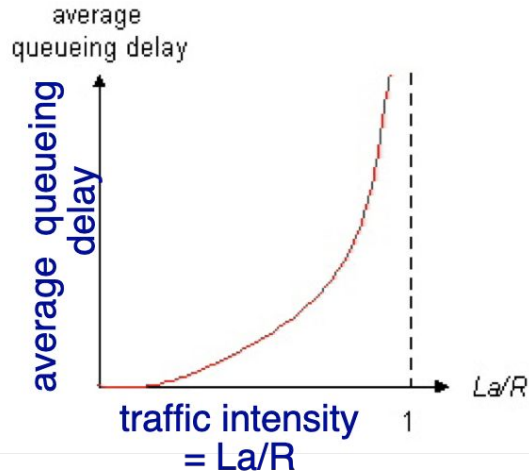
- The first bit reaches the receiver before the sender finishes transmission
- The data pipe is FULL

$$d_{\text{prop}} > d_{\text{trans}}$$

- Sender finishes transmission; receiver has not received even the first bit
- Typical scenario in case of long distance links

- **Limiting factor: How fast the link is?**
- **Design packet size for better network utilization (pipe should be FULL)**
- **Design protocols for better network utilization (pipe should be FULL)**

Queueing delay



R: Link bandwidth (bits per sec)

L: Packet length (bits)

a: average packet arrival rate
(num_pkts per sec)



$\frac{La}{R} \sim 0$



$\frac{La}{R} \rightarrow 1$

- $\frac{La}{R} \sim 0$
 - Small average queueing delay
- $\frac{La}{R} \rightarrow 1$
 - Large average queueing delay
- $\frac{La}{R} > 1$
 - Infinite average delay
 - More work; less capacity

average
queueing delay

average queueing
delay

traffic intensity
= $\lambda a / R$

1

$\lambda a / R$

How do “real” Internet delays and losses look like?

Traceroute tool

You will study “traceroute” and many more network tools in your tutorial & programming assignment.

Concept of end to end delays

Problem 3

Consider a link A--S--B. A and B are end hosts and S is a switch/router in between. Each link has length 200m (speed of light in medium is 2×10^8 m/s). Each link has bandwidth of 1 Mbps. "A" has sent 125 byte packet. Each node (switch/end-host) takes 9 μ s to process each received packet. What is the **end-to-end delay** for this 125 byte packet **in microseconds**?

Benefits of packetization in store-and-forward networks

Problem 4

Consider a link A--S--B problem (**problem 3**). Suppose “A” wants to send a 1MB file to “B”. “A” has two options:

- a. Send the entire file as one big packet on both the links
- b. Break up the packet into 100 byte chunks, add a 25 byte header per chunk, and send it on the links.

Ignore the propagation and processing delays, as they are very small.

Compute the **end-to-end delays** for both the cases.

Circuit switching vs. Packet switching delays

Problem 5

Consider a network with two end hosts and one switch in between, A--S--B. Each link has bandwidth of 1 Mbps. Suppose A and B wish to transfer a 1MB file between them. The switch can do circuit switching(CS) or packet switching(PS). **Ignore propagation delays.**

For CS: Setup request/response message = 1KB each; node processing delay = 2ms. After the circuit is setup, data is pumped at link rate.

For PS: Split a file into 100 byte segments and add 25 byte headers; **ignore node processing delays (~ 0)** $\rightarrow$ **otherwise, $t_{proc} > t_{trans}$** Use datagram approach of packet switching (i.e., No call setup involved)

1. What are the delays and overheads (in “num bytes”) for both, CS and PS?
2. What are the pros and cons of each?

Bandwidth Delay Product (BDP) calculation

Problem 6

A sender and receiver are communicating over a 1Mbps link. Sender sends **1KB packets**. Receiver sends **250 byte ACKs**. Suppose **round trip propagation delay** is **10 ms** (assume long distance link).

- A. What is the throughput if sender sends 1 packet and waits for an ACK?
- B. What is the bandwidth delay product (BDP) of the link?
- C. How is BDP useful for network design?